

**Title:**

Interpretable Attribution of Frontier Language Models Across Languages and Translation

**Abstract:**

Identifying the provenance of AI-generated text is increasingly important for tracing and understanding the growing volume of content produced by large language models. As these models generate text across more languages, however, attribution to a specific source model remains largely English-centric and often relies on closed or proprietary techniques, which limit interpretability and accessibility. We investigate whether interpretable stylometric features can distinguish five contemporary language models across seven languages representing different levels of estimated training-data availability. We evaluated 840 responses to semantically matched persuasive-writing prompts in English, Spanish, Chinese, Russian, Japanese, Turkish, and Hindi, extracting 21 language-comparable lexical, syntactic, structural, and punctuation features. A five-way classifier distinguished all five models in every language, achieving 59.2–71.7% accuracy versus 20% chance. Crucially, attribution accuracy did not significantly decline as estimated training-data availability decreased. At the same time, model-specific stylistic profiles became progressively more similar in lower-resource languages, with mean pairwise separation decreasing from 3.59 in English to 2.69 in Hindi ( $p = -0.96$ ,  $p < 0.001$ ). Despite this convergence, model-specific signals were transferred above chance across all tested language pairs and persisted through machine translation. In contrast, direct LLM-as-judge attribution remained near chance. These findings reveal that model-specific stylistic signatures persist across linguistic and translation boundaries even as the models themselves become stylistically more similar in lower-resource settings. More broadly, multilingual AI systems may simultaneously preserve identifiable traces of their source models while reducing stylistic diversity in lower-resource languages.

**Summary:**

AI-text detection tools are largely developed and evaluated in English, while the applications of determining where a text came from (provenance) are increasingly multilingual. Existing methods for attributing text to a specific model have focused on English and commonly use opaque embedding classifiers. We test whether interpretable stylometric features can identify source models across languages and whether attribution changes with training-data availability across languages, or translation.

We evaluated GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3, and DeepSeek V4 Pro on 12 controlled persuasive-essay prompts in English, Spanish, Chinese, Russian, Japanese, Turkish, and Hindi. Each prompt fixed the topic, stance, three supporting points, and reading level. The models generated two responses per prompt-language condition, yielding 840 responses. We extracted 21 stylometric features spanning lexical diversity, syntax, structure,

and punctuation. We used leave-one-prompt-out five-way classification with logistic regression as the primary interpretable model.

The classifier distinguished all five models in every language, achieving 71.7% accuracy in English, 62.5% in Spanish, 65.0% in Chinese, 65.0% in Russian, 70.8% in Japanese, 59.2% in Turkish, and 63.9% in Hindi, compared with 20% chance. Attribution did not significantly decline with decreasing training-data share (Spearman  $\rho = -0.45$ ,  $p = 0.31$ ; cell-level GLM  $\beta = -0.038$ ,  $p = 0.36$ ). Language explained more feature variance than model identity (mean partial  $\eta^2 = 0.54$  vs.  $0.15$ ), yet cross-lingual transfer was above chance for all 42 language pairs, averaging 0.49 versus 0.65 within language. The separation between model-specific feature profiles nevertheless converged as resources fell, with mean pairwise distance decreasing from 3.59 in English to 2.69 in Hindi ( $\rho = -0.96$ ,  $p < 0.001$ ). Paragraph count, comma and colon density, and lexical-diversity measures retained model-specific signals.

Surprisingly, translation also preserved the fingerprint. Across 1,440 translations, attribution averaged 0.625 for Google Translate and 0.629 for the LLM translator, versus 0.717 for English originals and 0.644 for native responses. An English-trained classifier achieved 0.66 on translated text. Structural features survived strongly, including paragraph count ( $p = 1.00$ ) and sentence length ( $p = 0.89$ ), while lexical measures changed more substantially.

As a baseline comparison, direct LLM-as-judge attribution was near chance: GPT 20.6%, Grok 21.9%, Gemini 21.2%, DeepSeek 20.3%, and Claude 25.4%. Claude was the only judge slightly above chance. The interpretable classifier exceeded every judge by 35–50 percentage points in every language. These results show that model-specific stylometric fingerprints remain measurable across the tested resource gradient and translation boundary, while frontier models converge stylistically in low-resource languages. Code, prompts, all responses, and translations are publicly released.